

The empirical recurrence for OEIS A209105, proved by transfer matrix

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Abstract

OEIS A209105 counts $6 \times n$ arrays over $\{0, \dots, 2\}$ subject to a condition imposed on every CELL over a neighbour set the entry names, and counted UP TO RELABELLING of the alphabet; it carries an empirical recurrence of order 72 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. What is counted is equality patterns, which is not a walk count — but it is a fixed rational combination of walk counts, one for each alphabet size, and that combination is again C -finite. Whether the conjectured recurrence annihilates it is then decided by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

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1 The conjecture

OEIS A209105 is “Number of $6 \times n$ 0..2 arrays with new values 0..2 introduced in row major order and no element equal to more than one of its immediate leftward or upward or right-upward antidiagonal neighbors.”. It has offset 1 and begins

122, 34096, 7154944, 1404480904, 272236743760, 52675800891748, 10191444894367900, 1971869773009191300, . .

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = 483*a(n-1) - 93045*a(n-2) + 9895994*a(n-3) - 655000699*a(n-4) + 28153218507*a(n-5) - 771472287050*a(n-6) + 11575521770914*a(n-7) - 5333969777492*a(n-8) - 3460511755733129*a(n-9) + 60619944832280219*a(n-10) - 122454879122841094*a(n-11) - 9610060024477677636*a(n-12) + 119490257201622127599*a(n-13) + 207423246569123218904*a(n-14) - 15418046735278157348540*a(n-15) + 85958015930994934073828*a(n-16) + 809517273485108604329443*a(n-17) - 10872862333905128370604528*a(n-18) - 553924361897541639230341*a(n-19) + 628503974578083970787751095*a(n-20) - 2441837003902470322705776546*a(n-21) - 19135804539591412466515125841*a(n-22) + 157196371353874391098869608556*a(n-23) + 184930685928293621578721897714*a(n-24) - 5212791777478985863598283828640*a(n-25) + 8343380811716464763432941624104*a(n-26) + 99943078075370200668702342185854*a(n-27) - 385486750658814649071993027833855*a(n-28) - 947326129545439583693615334446414*a(n-29) + 7617076417557384976850188004727513*a(n-30) - 1587365613230665275472648686777016*a(n-31) - 81893089783126645169411368168862676*a(n-32) + 154905729052219569334495329758007735*a(n-33) + 392446436386226179411197434377941811*a(n-34) - 1789366396909098001371959030444799187*a(n-35) + 1364382304774676286186085515863856887*a(n-36) + 7450756681407735007607770239017405077*a(n-37) - 32742420723798992453476401619165532832*a(n-38) + 34882707064638274850232798681677400146*a(n-39) + 196690238938567903829556617524922100465*a(n-40) - 642195656452343483207376794444201641353$

41) -302741845834579655279069984112105879663*a(n-42) +389893071193634836996280524357031823433
 43) -2811628196998770649303159460561101126472*a(n-44) -126991543077377938696619850768534261555
 45) +20357131154883109280177933384191456925974*a(n-46) +206438835390662332740840306268036967
 47) -64755513215769597861644458739579161901156*a(n-48) -92817721136347903374524011401255653970
 49) +116393013534805718816923463675254988333020*a(n-50) -629227646134740312775054085205219401
 51) -117636718952533538147029798845965591298928*a(n-52) +122662194993078191268927575331077554
 53) +53377954500120792927391233551189754963712*a(n-54) -1116601323973036898293568514212520482
 55) +7919142148317395919389979426042637954304*a(n-56) +5151840982743894732195421434130627866
 57) -18670349951681657511291462879827883996160*a(n-58) -10933253490659848148544920316312898695
 59) +6995956832033163694328578109174268706816*a(n-60) +1686799976967671987881917375839533137
 61) -1253524043678855280232828496400957308928*a(n-62) -906202586627792174825114955918683930624
 63) +309998526763988330640199202118903529472*a(n-64) +14855262282182719264409243687478047539
 65) -20331273732734054176046805671907164160*a(n-66) +93663781756651920952176320882940051456*a
 67) -77028642178807633010626944466396643328*a(n-68) +1689650677968769324222545429484535808*a(n-
 69) +8972893867255780734625567627712397312*a(n-70) -855929165188635245245315361739374592*a(n-
 71) -26033153048978658111126222776041472*a(n-72) for n>76

As of the “Last modified” line on the live entry (Oct 03 2025 15:09:24, revision 9) this is still recorded as empirical, and nothing on the entry records it as proved.

2 Patterns are a combination of counts

The clause “new values 0..2 introduced in row major order” says that the arrays counted are the canonical representatives of their EQUALITY PATTERNS: two arrays that differ only by a permutation of the alphabet are one object. Write N_j for the number of admissible patterns using exactly j letters and L_i for the number of admissible ARRAYS over an alphabet of i letters. An array over i letters is a pattern together with an injection of its letters into the alphabet, so

$$L_i = \sum_j N_j i(i-1) \cdots (i-j+1),$$

a triangular system with nonzero diagonal, hence invertible over $\mathbb{Q}$. The entry counts $a = N_1 + \cdots + N_3$, so

$$a = (\mathbf{1}^\top F^{-1})L,$$

a FIXED rational combination of the L_i , where F is the matrix of falling factorials. Each L_i is a walk count on its own transfer graph, so a is a walk count on the disjoint union of those graphs with the combination’s coefficients placed in the starting vector; the coefficients are rational and some are negative, which costs nothing in what follows.

This step needs the condition itself to be unchanged by permuting the alphabet, and it is: the condition below is stated through equality between entries and never names a particular value or compares sizes.

3 Cells, not blocks

The entry names the neighbours it means — horizontal is left and right, vertical is up and down, diagonal is northwest-to-southeast, antidiagonal is northeast-to-southwest, king-move is all eight — and every such set lies inside the 3×3 square centred on the cell. Here the set is $\{(-1, 0), (0, -1), (1, -1)\}$, written as offsets (row, column) from the cell. Every cell carries the condition

no cell is equal to more than 1 of its immediate leftward or upward or right-upward antidiagonal neighbours,

and the count of neighbours enters it, so the boundary is not a detail that can be papered over: a cell in the interior has 3 neighbours in this set and a cell on the boundary fewer, and a condition such as “a strict majority of its neighbours” therefore means different things at the two. In particular one cannot pad the array with a border of zeros — that would give every cell the full 3 and change what is being counted. The outside is therefore carried explicitly, as a sentinel column above the first and below the last and as a sentinel column either side, and a neighbour that falls outside is simply absent from both counts.

Write an array as a sequence of columns $u_1, \dots, u_L$, each in $\{0, \dots, 2\}^6$. The neighbourhood of a cell in column u_i meets u_{i-1} , u_i and u_{i+1} and nothing else. Take as vertices the pairs $(r, s) \in (\{0, \dots, 2\}^6 \cup \{\emptyset\}) \times \{0, \dots, 2\}^6$, and put an edge $(r, s) \rightarrow (s, t)$ when column s satisfies the condition at every one of its 6 cells, given r above and t below. An array with L columns is then a walk of length $L - 1$ starting at $(\emptyset, u_1)$, each step settling one column, and the last column is settled by the terminal weight, which evaluates it with $\emptyset$ below. Hence, with M the adjacency matrix, ι the starting vector and τ the terminal weight,

$$a(n) = \frac{1}{1} \iota^\top M^{n-1} \tau,$$

the exponent fixed by the entry’s own shape and checked against its published terms. In particular a is C -finite.

4 The proof

Lemma 1. *Let $q(t) = t^{72} - \sum_i c_i t^{72-i}$ be the characteristic polynomial of the conjectured recurrence. Then the residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j ; and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+72} - \sum_i c_i M^{j+72-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ satisfy the monic linear recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 2.

$$a(n) = 483a(n-1) - 93045a(n-2) + 9895994a(n-3) - 655000699a(n-4) + 28153218507a(n-5) - 771472287050$$

for every $n > 76$.

Proof. The vector $w = q(M) \tau$ was formed by 72 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly. Those integers vanish from the index corresponding to $n = 76 + 1$ onwards, and the run was continued until the working vector was identically zero, which settles all larger j at once. By Lemma 1 the recurrence holds for every $n > 76$. $\square$

5 Verification

Three checks, each independent of the algebra above.

First, the transfer model reproduces all 11 terms the entry publishes, exactly, in integer arithmetic. That check earned its place here. The neighbour offsets are named in the array’s own frame — horizontal means along a row — but when the entry fixes the first dimension, as in “ $2 \times n$ ”, the walk runs along columns and the two components of every offset swap. Sets such as king-move, or horizontal together with vertical, are unchanged by that swap and hide the error

completely; a set such as horizontal, diagonal and antidiagonal is not, and the counts came out as those of a different member of the same family until the offsets were transposed.

Second, the conjectured recurrence was evaluated directly on the entry's own published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.